

```
BEGIN ISOLATION LEVEL REPEATABLE READ;
```

```
SELECT * FROM accounts WHERE id = 1;
```

```
-- More queries
```

```
COMMIT;
```

or equivalently:

```
BEGIN;
```

```
SET TRANSACTION ISOLATION LEVEL REPEATABLE READ;
```

```
SELECT * FROM accounts WHERE id = 1;
```

```
COMMIT;
```

What does REPEATABLE READ do?

It guarantees that all reads inside the transaction see the same

snapshot of data.

Example

Session 1

```
BEGIN ISOLATION LEVEL REPEATABLE READ;
```

```
SELECT balance  
FROM accounts  
WHERE id = 1;
```

Result:

1000

Session 2

```
UPDATE accounts  
SET balance = 2000
```

```
WHERE id = 1;
```

```
COMMIT;
```

Session 1

```
SELECT balance  
FROM accounts  
WHERE id = 1;
```

Result:

1000

Even though Session 2 committed the change, Session 1 still sees the old value because it is using the same snapshot taken when the transaction started.

After Session 1 commits and starts a new transaction:

```
COMMIT;
```

```
BEGIN;
```

```
SELECT balance  
FROM accounts  
WHERE id = 1;
```

Result:

2000

READ COMMITTED → See only committed data, refreshed for every query.
(default)

REPEATABLE READ → See the same data snapshot throughout the transaction. even if other transactions commit changes it won't show.

SERIALIZABLE → Prevents all concurrency anomalies by making transactions appear sequential.

Example:

```
BEGIN ISOLATION LEVEL SERIALIZABLE;
```

```
BEGIN ISOLATION LEVEL REPEATABLE READ;
```

```
BEGIN ISOLATION LEVEL READ COMMITTED;
```